

ST

MARS

THE PRISTINE BEAUTY OF THE RED PLANET

ALFRED S. MCEWEN • CANDICE HANSEN-KOHARCHECK • ARI ESPINOZA

INTRODUCTION

The Mars Reconnaissance Orbiter (MRO) launched from Florida on August 12, 2005, carrying the High Resolution Imaging Science Experiment (HiRISE), the most powerful camera ever sent to another planet (by Earthlings). Our team was first selected to build HiRISE in November 2001, less than four years before launch. That was a very fast schedule for such a complex instrument!

No camera like this had ever been built before. It is similar in terms of resolution and swath width to the QuickBird imager, launched into Earth orbit on October 18, 2001. However, the HiRISE instrument needed to have only one-fifth the weight (or mass) of the QuickBird imager because it is much more expensive to place mass in Mars' orbit than in Earth's orbit. This lightweighting required a new design, a major effort before we could actually build the instrument. Both the QuickBird and HiRISE instruments were designed and built by Ball Aerospace in Boulder, Colorado, which did a magnificent job. (For more about the MRO mission, see <http://mars.nasa.gov/mro/>.)

NASA's requirement for MRO was high-resolution (approximately 0.3 cm/pixel) black-and-white imaging. However, we thought it was important to add detectors to provide color imaging. HiRISE images are about five kilometers wide in a broad red band-pass (from 10 CCDs), with a one-kilometer central stripe in three colors from the addition of four CCDs, two with a blue-green band-pass and two with a near-infrared band-pass. This very high-resolution color imaging has proven to be extremely valuable to HiRISE science and landing site reconnaissance. More importantly for this book, the color images are beautiful!

HiRISE does not see the same colors as normal human vision, analogous to having blue, green, and red band-passes. Instead, HiRISE color is shifted to longer wavelengths, similar to many Landsat images in which green vegetation appears red. This shift helps us to better see surface features through the dusty atmosphere of Mars and to better distinguish rock types from their mineral content. Most of the HiRISE images in this volume are IRB color, with the infrared channel displayed as red, the broad red channel as green, and the blue-green channel as blue. In addition, we produce a color product that uses only the red and blue-green channels because one of the infrared detectors is often too noisy to produce useful images. We call these RGB images, and they are closer to, but still do not match, normal human color vision. RGB images display the broad red channel as red, the blue-green channel as green, and a synthetic blue channel as blue. Subtracting a fraction of the red channel intensity from the blue-green channel produces the synthetic blue channel. Both the IRB and RGB images are "stretched" to improve contrast. For example, if the darkest pixel value is 2,000 and the brightest is 4,000, we set 2,000 to black and 4,000 to white and linearly interpolate the intermediate values. If we did not stretch the images, they would usually be very bland because about half of the top-of-atmosphere signal is scattered light from the dusty air.

MRO is in a Sun-synchronous orbit that passes over the dayside of Mars at approximately 3 p.m. on each of 12.8 orbits per day. The middle of the afternoon is the best time to image the surface

because there is generally less atmospheric haze than in the morning and we want an intermediate illumination angle—high enough to accentuate the topography while still providing plenty of light. Standard images in this volume have north up and the illumination coming from the west (left side). However, we deviate from that standard (and say so in the captions) for some images. Sometimes we rotate the image so north is to the left or right to display a larger image over two pages. Sometimes we show an image with north down, in particular landscapes that slope downward that way, which is more intuitive. Scale bars are not provided, to keep the images uncluttered, but the image scenes are typically about one kilometer (0.6 mile) wide.

Also included throughout the book are topographic images derived from stereo pairs. These are color coded to relative altitude (white and red are high areas, and blue and purple are low) and then merged with the black-and-white image to show surface details. North is up on all of these images, and the illumination comes from the west (left).

Image identifiers are provided for all images. If you want to see the whole image and all data products, go to our website. For example, to see image ESP_011523_1695, go to http://uahirise.org/ESP_011523_1695. Our website provides hundreds of additional captioned images in more than 20 languages. Captions were initially written by over 30 different HiRISE team members, which helped to ensure that the most interesting images were included. The authors edited these captions to provide a degree of uniformity, but the readers may notice some different styles. Informal names for places on Mars, not approved by the International Astronomical Union, are put in quotations, for example “Columbia Hills.” Finally, we have tried to minimize and to define technical jargon, but we’ve also included a glossary for those terms we could not avoid using.

As of August 2016, HiRISE has acquired over 45,700 images of Mars, and these are large images, typically $20,000 \times 40,000$ pixels. MRO has returned far more data than any other mission to another planet, and about 40 percent of that has been HiRISE data. In spite of this great volume of data, HiRISE has imaged only about 2 percent of the Martian surface. About 2.6 percent of the surface would be covered if it was all unique, but we acquire a lot of repeat coverage to track changes over time and for “stereo images” (separate images looking from the left and right). HiRISE has imaged less than 0.5 percent of Mars in color. Mars has about the same surface area as the land area of Earth, and at 30 cm/pixel scale, that is too much data to send from Mars to Earth via traditional radio transmissions. NASA is working toward optical communication technology to return significantly greater volumes of data from Mars, so a HiRISE-like instrument on a future orbiter, hopefully with the full swath width in color, could return much greater coverage of the planet.

What is the future of HiRISE? The MRO spacecraft has enough fuel to continue operations for another 20 years, but it is not likely to last that long. Mechanical and electronic parts fail eventually, and MRO is already well past its design life. We cannot predict the future, but we hope HiRISE will complete a second decade.

Alfred S. McEwen
Candice Hansen-Koharcheck

There is a vast and cool silence on Mars, sometimes broken by the sound of a boulder perched at the rim of an ancient impact crater—formed long before the Code of Hammurabi or Plato's tale of Atlantis—giving way to inexorable gravity, finally rolling down a slope before coming to a stop and continuing its vigil. Somewhere at the Martian poles, warmed by weak but still potent heat from the sun, ice cracks on a cliff, sending material plummeting below, raising dust clouds that linger and slowly disappear. The wind works relentlessly, yes, its echo everywhere across sculpted rocks, but mostly, this is a quiet world.

Except no one would have believed that this eerie silence would be broken by a new sound, a screaming coming across the sky, the alien noise of machines sent from another planet. Yet these are not war machines sent on a missionary enterprise; they are machines of exploration, of intellectual and scientific curiosity, whose important data are pored over by keen eyes, adding new layers of knowledge about a planet that has captured our imagination for so long. Ironically, our machines may be lasting testaments to those who sent them, whether on the ground or in orbit, as instruments of peace and discovery, even if a human foot never takes the next giant leap there.

Space exploration needs engineers, physicists, mathematicians, geologists, and astronomers. But it also needs poets, thinkers, and dreamers. We need people to help explain why we explore in the first place: not because it is easy but because it is hard, as John F. Kennedy said of going to the Moon. With this book, we present to you a snapshot of our current scientific knowledge of Mars, coupled with the most stunning images of the surface of another planet ever taken. The detail is astonishing, unprecedented. The frozen canvas that is Mars is so powerful on its own that we easily could have created a volume with little interpretation, letting the pictures stare back at you. But that would not work. In keeping with the philosophy of HiRISE being "the people's camera," we wanted to do something more, something to make these artistic landscapes move and come alive. If anything, this is the most original science book ever made, bringing voices from across our world together to describe another one, in languages that many people know and in others that have never before been used in a single volume. This book is our version of the Golden Record.

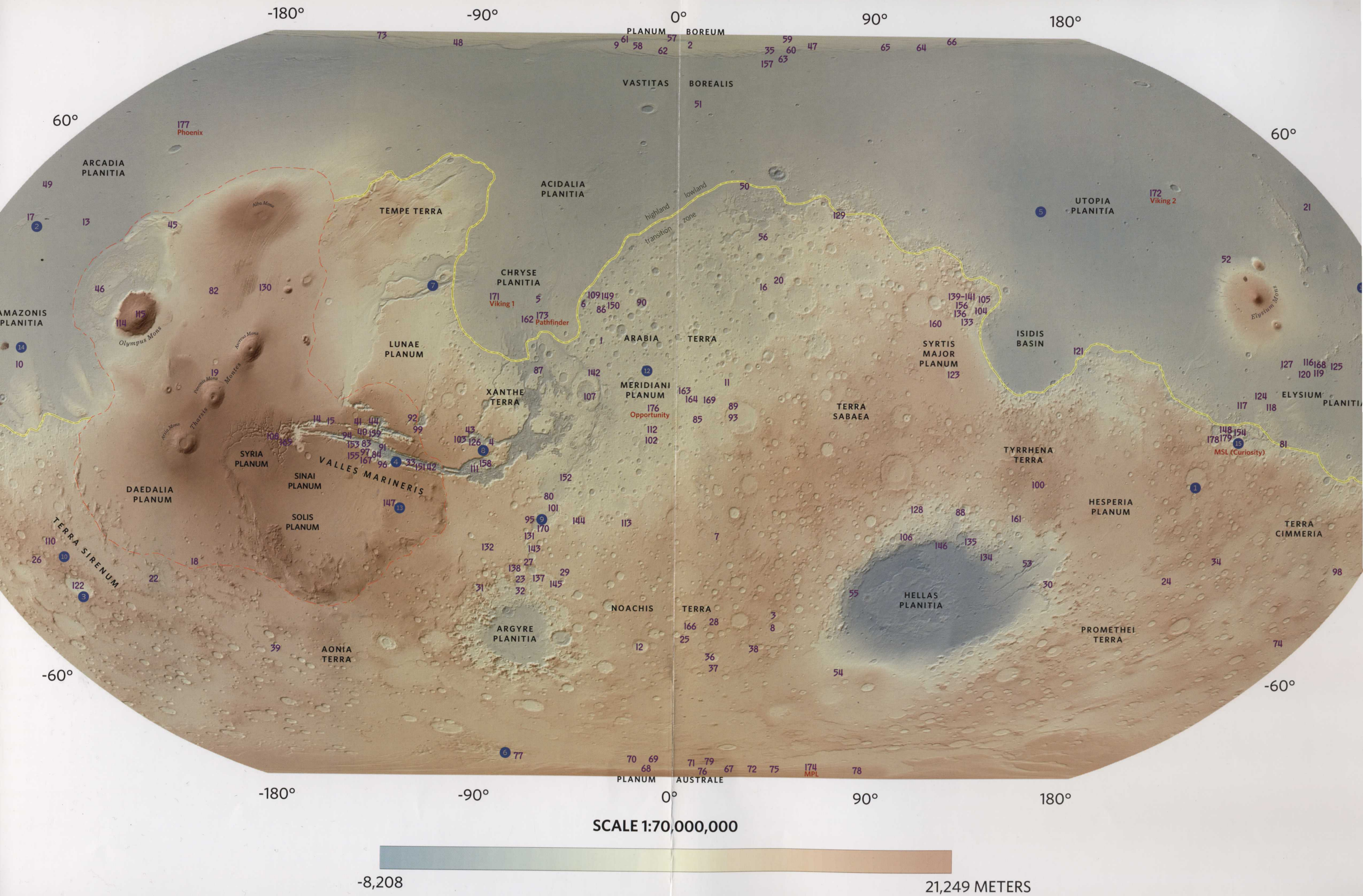
So Mars remains quiet and unmoving, but it is our wish that this work, the product of so many different and talented individuals, will inspire you to add your own voice to the images that we are proud to offer.

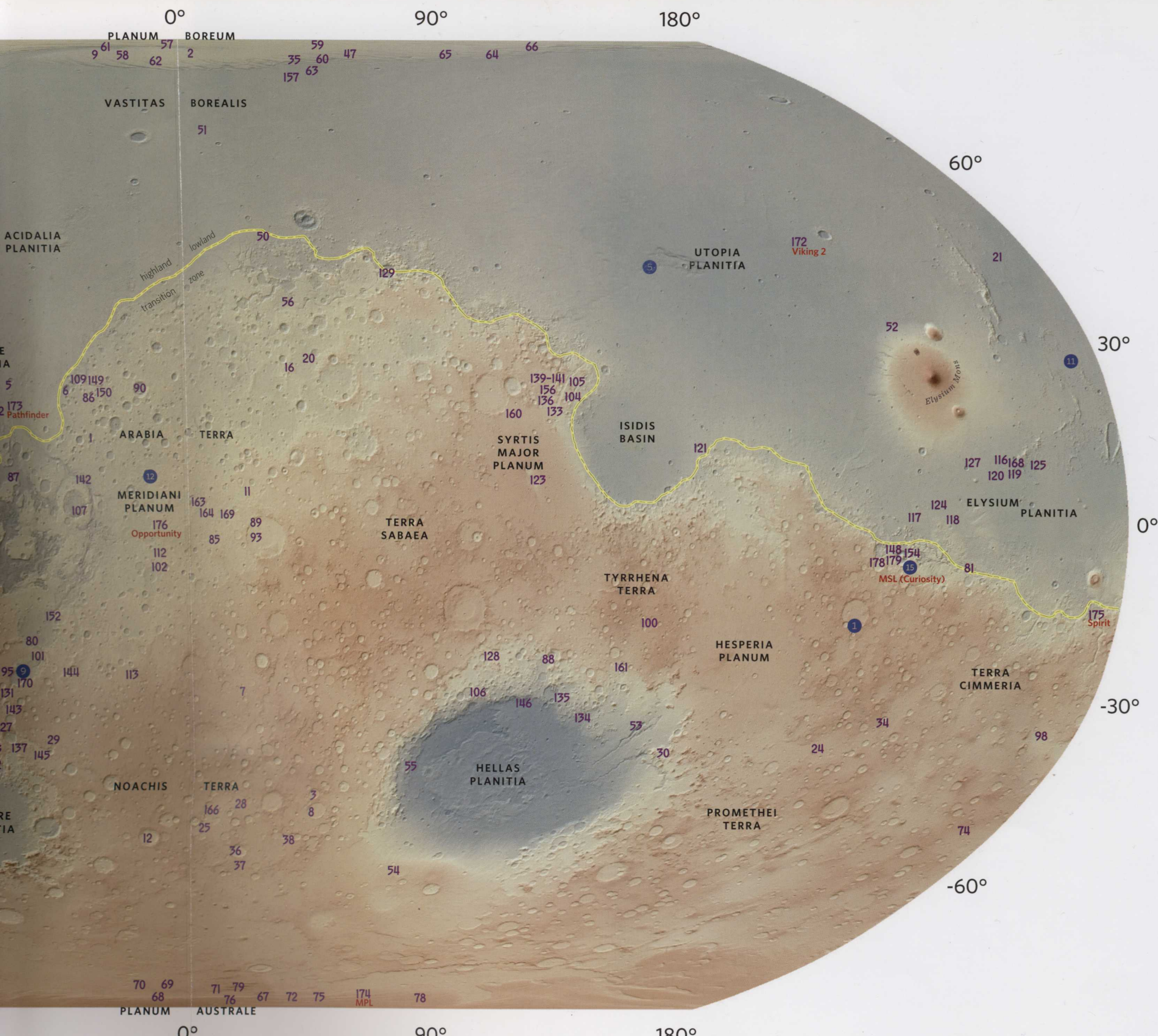
Ari Espinoza

Map of Images as They Appear in This Volume

(digital terrain models are indicated with circled numbers)

1. ESP_014429_1940	p. 5	46. ESP_014394_2045	p. 106	91. ESP_026312_1700	p. 216	136. PSP_009573_2035	p. 325	Digital Terrain Models	
2. ESP_019161_2650	p. 6	47. ESP_016228_2650	p. 109	92. PSP_003579_1755	p. 219	137. ESP_020166_1440	p. 327	1. PSP_002860_1650	p. 30
3. ESP_022181_1345	p. 8	48. ESP_016423_2640	p. 111	93. ESP_017118_1790	p. 221	138. PSP_008615_1445	p. 329	2. ESP_025735_2185	p. 50
4. ESP_019520_1725	p. 11	49. PSP_010861_2265	p. 116	94. PSP_005597_1740	p. 222	139. PSP_002888_2025	p. 335	3. PSP_005943_1380	p. 90
5. ESP_017647_2025	p. 13	50. ESP_018885_2270	p. 119	95. ESP_011542_1530	p. 225	140. PSP_008782_2015	p. 337	4. ESP_020879_1670	p. 112
6. PSP_009814_2020	p. 14	51. ESP_017131_2485	p. 120	96. ESP_013719_1665	p. 226	141. PSP_006923_1995	p. 339	5. ESP_025277_2275	p. 132
7. ESP_041276_1515	p. 17	52. PSP_005879_2150	p. 122	97. PSP_007667_1700	p. 229	142. ESP_013216_1870	p. 340	6. ESP_022699_0985	p. 184
8. ESP_021469_1325	p. 19	53. PSP_008597_1460	p. 125	98. ESP_012708_1450	p. 231	143. ESP_013586_1490	p. 343	7. PSP_002814_2055	p. 209
9. PSP_009682_2630	p. 20	54. PSP_004340_1235	p. 127	99. PSP_009842_1755	p. 233	144. PSP_009089_1550	p. 345	8. ESP_017173_1715	p. 243
10. ESP_021568_1925	p. 22	55. ESP_025635_1395	p. 129	100. PSP_001503_1645	p. 234	145. ESP_012821_1420	p. 346	9. PSP_001336_1560	p. 244
11. ESP_022300_1900	p. 25	56. PSP_009588_2175	p. 131	101. PSP_006637_1590	p. 237	146. ESP_016536_1505	p. 349	10. PSP_009318_1465	p. 268
12. PSP_005397_1270	p. 26	57. PSP_010545_2700	p. 137	102. ESP_011620_1750	p. 239	147. PSP_003408_1585	p. 351	11. ESP_018747_2065	p. 294
13. ESP_026051_2160	p. 29	58. PSP_001334_2645	p. 139	103. ESP_013059_1725	p. 240	148. ESP_019988_1750	p. 353	12. PSP_002878_1880	p. 310
14. ESP_011425_1775	p. 35	59. PSP_010716_2640	p. 140	104. PSP_009362_2020	p. 249	149. PSP_006610_2035	p. 354	13. PSP_002353_1585	p. 330
15. ESP_012282_1775	p. 37	60. PSP_009599_2615	p. 143	105. ESP_016865_2035	p. 250	150. ESP_013295_2050	p. 356	14. PSP_003543_1910	p. 385
16. PSP_003101_2065	p. 39	61. ESP_018852_2655	p. 144	106. ESP_014189_1525	p. 253	151. ESP_011543_1665	p. 359	15. PSP_009149_1750	p. 424
17. ESP_016083_2205	p. 40	62. PSP_009166_2650	p. 147	107. ESP_012860_1830	p. 255	152. ESP_011634_1645	p. 360		
18. PSP_005045_1455	p. 43	63. PSP_008530_2505	p. 148	108. ESP_016120_1730	p. 256	153. TRA_000823_1720	p. 367		
19. PSP_003172_1870	p. 45	64. ESP_019143_2635	p. 151	109. ESP_022354_2035	p. 258	154. PSP_007501_1750	p. 368		
20. ESP_019926_2070	p. 47	65. PSP_010086_2615	p. 153	110. ESP_013274_1495	p. 261	155. PSP_003118_1690	p. 371		
21. PSP_010189_2265	p. 49	66. ESP_018760_2650	p. 155	111. ESP_019652_1665	p. 262	156. PSP_008993_2020	p. 372		
22. TRA_000878_1410	p. 54	67. ESP_014345_1070	p. 156	112. ESP_016354_1745	p. 265	157. PSP_009706_2555	p. 375		
23. ESP_016184_1410	p. 57	68. ESP_020734_0930	p. 159	113. ESP_013018_1550	p. 267	158. ESP_022632_1670	p. 376		
24. ESP_022217_1440	p. 59	69. PSP_005728_0935	p. 160	114. PSP_008526_1980	p. 273	159. PSP_001984_1735	p. 378		
25. ESP_012266_1300	p. 60	70. ESP_011610_0935	p. 163	115. PSP_007023_1980	p. 274	160. PSP_001385_1985	p. 381		
26. ESP_011969_1425	p. 63	71. ESP_020678_0930	p. 167	116. PSP_007197_1910	p. 277	161. ESP_022324_1555	p. 382		
27. ESP_012874_1450	p. 65	72. ESP_011842_0980	p. 168	117. ESP_013830_1845	p. 278	162. ESP_021748_1990	p. 388		
28. ESP_022406_1335	p. 66	73. ESP_017056_2640	p. 171	118. PSP_008529_1855	p. 281	163. PSP_003735_1825	p. 391		
29. ESP_022315_1440	p. 69	74. PSP_009781_1295	p. 173	119. PSP_001408_1900	p. 283	164. PSP_003972_1805	p. 392		
30. ESP_016258_1415	p. 70	75. ESP_012081_0930	p. 175	120. PSP_001606_1900	p. 284	165. PSP_010397_1725	p. 394		
31. ESP_022540_1405	p. 73	76. ESP_011960_0925	p. 177	121. ESP_014016_1945	p. 287	166. PSP_004091_1325	p. 397		
32. ESP_011938_1410	p. 74	77. ESP_011729_0985	p. 178	122. PSP_005877_1375	p. 289	167. ESP_019508_1700	p. 398		
33. ESP_042228_1670	p. 77	78. ESP_011749_1000	p. 181	123. ESP_013371_1875	p. 290	168. PSP_009280_1905	p. 401		
34. PSP_005787_1475	p. 79	79. ESP_011370_0980	p. 183	124. ESP_020581_1835	p. 293	169. ESP_012226_1815	p. 402		
35. PSP_007962_2635	p. 81	80. ESP_011476_1600	p. 190	125. ESP_017719_1890	p. 299	170. ESP_019256_1530	p. 405		
36. ESP_020217_1255	p. 83	81. ESP_016269_1750	p. 193	126. PSP_003684_1730	p. 301				
37. ESP_022340_1255	p. 85	82. PSP_002592_2030	p. 194	127. PSP_004191_1890	p. 302	171. PSP_001521_2025 (Viking 1)	p. 408		
38. ESP_021733_1275	p. 87	83. ESP_013561_1705	p. 197	128. ESP_012989_1585	p. 305	172. PSP_001976_2280 (Viking 2)	p. 409		
39. ESP_011400_1275	p. 88	84. ESP_020286_1700	p. 199	129. ESP_017710_2215	p. 307	173. PSP_002391_1995 (Pathfinder)	p. 411		
40. ESP_016185_1740	p. 95	85. PSP_006939_1780	p. 200	130. ESP_011386_2065	p. 309	174. ESP_013790_1035 (MPL)	p. 412		
41. PSP_005886_1715	p. 97	86. PSP_006755_2030	p. 203	131. ESP_011608_1525	p. 315	175. ESP_021925_1650 (Spirit)	p. 415		
42. PSP_008141_1645	p. 98	87. PSP_001415_1875	p. 205	132. ESP_022619_1495	p. 316	176. PSP_004289_1780 (Opportunity)	p. 416		
43. ESP_013217_1715	p. 101	88. ESP_013226_1580	p. 206	133. ESP_011538_1970	p. 319	177. PSP_008579_9020 (Phoenix)	p. 418		
44. ESP_011372_1730	p. 102	89. PSP_002930_1800	p. 213	134. ESP_022245_1485	p. 320	178. ESP_028256_9022 (Curiosity descent)	p. 421		
45. PSP_001656_2175	p. 105	90. PSP_004078_2015	p. 215	135. ESP_019186_1525	p. 323	179. ESP_032436_1755 (Curiosity)	p. 423		





SCALE 1:70,000,000

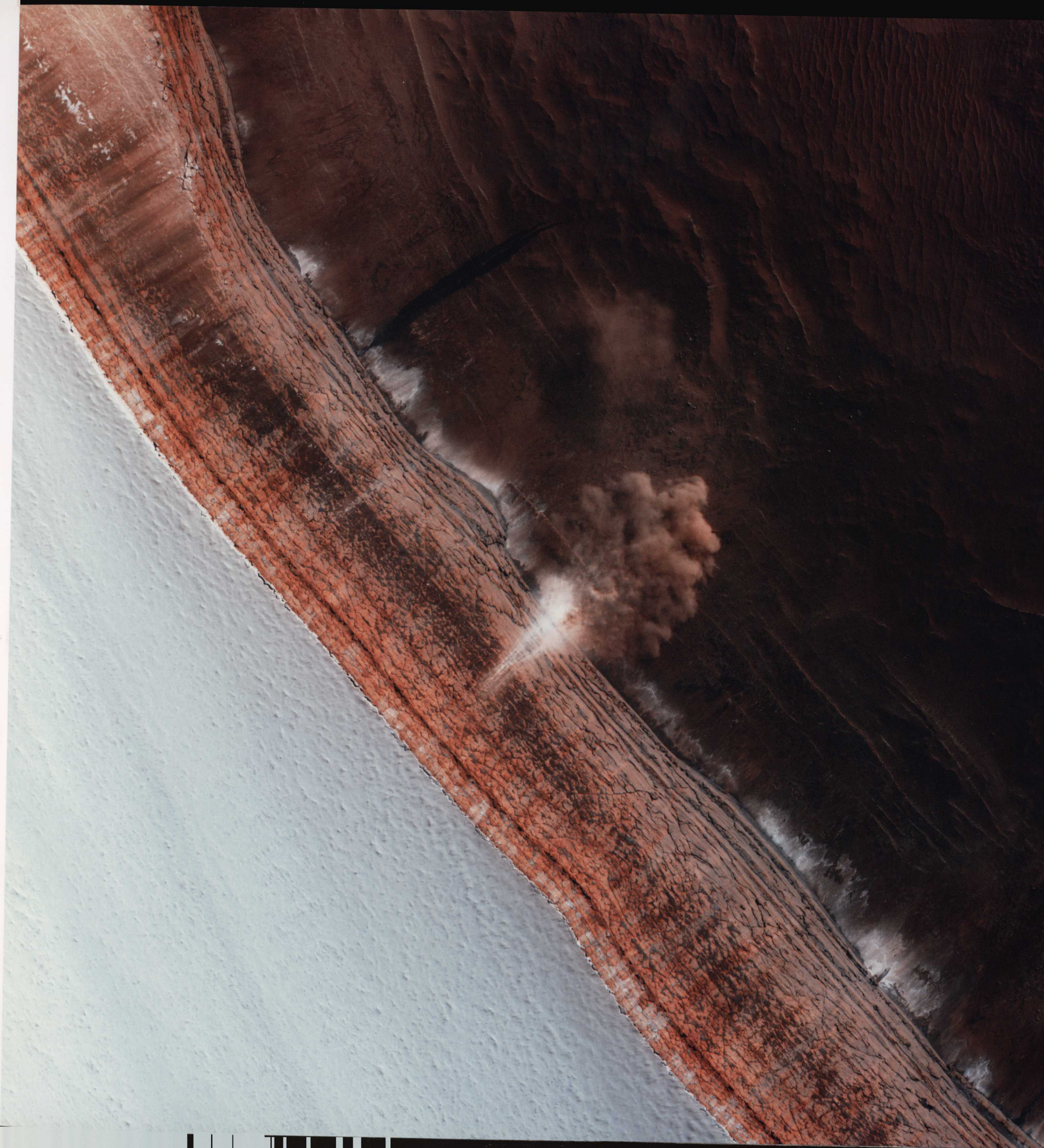
21,249 METERS

Scarps at the edge of the North Polar layered deposits are the sites of the most frequent frost avalanches that we've imaged. In the northern spring, frost avalanches are common, and we monitor these scarps to learn more about their timing and frequency and about how they relate to the evolution of frost on the flat ground above and below. The avalanches are likely carbon dioxide frost dislodged from the layers, and we can see material cascading down the cliffs, kicking up clouds of dust when they hit the ground. (ESP_016228_2650; see following page, ESP_016423_2640, for an image covering a larger area to provide regional context.)

Обрывы на краях слоистых отложений Северного полюса—места наиболее частых ледяных лавин, которые мы когда-либо фотографировали. Ледяные лавины обычны северной весной. Мы наблюдаем за этими обрывами, чтобы узнать больше о времени и частоте появления лавин и их связи с эволюцией льда выше и ниже по склону. Лавины—это, скорее всего, лед из углекислого газа, сорвавшийся с краёв слоистых отложений. Мы можем видеть этот материал, падающий со скал вниз и поднимающий облака пыли при столкновении с поверхностью.

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и столкновении с поверхностью.



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500 meters

3757 m



-1760 m

NASA/JPL/University of Arizona

MRO/HIRISE

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"Mars is one of the best places to search for signs of life beyond Earth, and it is a world that humans may one day call home. It is also a world of staggering beauty. The HiRISE camera on NASA's Mars Reconnaissance Orbiter has revealed the stark and sometimes bizarre landscapes of Mars in spectacular high resolution and vivid color for the first time. In *Mars: The Pristine Beauty of the Red Planet*, the best of those images are gathered together, providing the most dramatic views of the Red Planet ever seen. It's a book that belongs on the shelves of anyone who cares about Mars's ancient past, or about its role in humanity's future."

—STEVE SQUYRES, Principal Investigator, Mars Exploration Rovers



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